



M-P(MD365 F&O) (Middleware Integration) Design

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Summary:

The attached draft defines a two-step middleware integration between the external system and Microsoft Dynamics 365 Finance & Operations for customer and vendor payments:

1. Import payment journal lines.
2. Import and apply settlement against invoices.
Below is a cleaned, structured, implementation-ready version suitable for technical design or sharing with the middleware team.

Draft Middleware settlement

Middleware Integration with D365 F&O

Scope

Integration covers:

- Customer payment journals (Cash-in). General Journal Import Petty Cash
- Vendor payment journals (Cash-out). General Journal Import Petty Cash
- Automatic settlement against invoices.

The process is executed in **two steps** for both customers and vendors.

1) Customer Payment Integration (Cash-In)

Step 1: Import Customer Payment Journal

Middleware sends payment journal lines to D365.

Key Mapping (Mandatory Fields)

D365 Field	Source (Middleware)	Notes
JOURNALBATCHNUMBER	Middleware	Generated by middleware
LINENUMBER	Middleware	Generated by middleware
ACCOUNTDISPLAYVALUE	Account	Customer account
ACCOUNTTYPE	Account type	Customer

D365 Field	Source (Middleware)	Notes
COMPANY	Legal entity	Based on entity mapping
CURRENCYCODE	Currency	From source
DEBITAMOUNT / CREDITAMOUNT	Amount	Based on transaction
EXCHANGERATE	Exchange rate	From source
TRANSACTIONDATE	TransDate	Payment date
TRANSACTIONTEXT	Description	Default or from source
POSTINGPROFILE	Posting profile	Based on account type
MARKEDINVOICE	Invoice number	From same UniqueId
Payment Line (For Dynamics)	CashIn Excel	Notes
JOURNALBATCHNUMBER	Middleware	Created by middleware
LINENUMBER	Middleware	Created by middleware
ACCOUNTDISPLAYVALUE	ACCOUNTDISPLAYVALUE	
ACCOUNTTYPE	ACCOUNTTYPE	
BANKTRANSACTIONTYPE		Optional
CALCULATEWITHHOLDINGTAX	ISWITHHOLDINGCALCULATIONENABLED	
CENTRALBANKIMPORTDATE		Optional
CENTRALBANKPURPOSECODE		Optional
CENTRALBANKPURPOSETEXT		Optional
COMPANY		M-P(Based on legal entity)
CREDITAMOUNT	CREDITAMOUNT	
CURRENCYCODE	CURRENCYCODE	
CUSTOMERNAME	Middleware	Get from invoices(Dynamic)

D365 Field	Source (Middleware)	Notes
DEBITAMOUNT	DEBITAMOUNT	
DEFAULTDIMENSIONSFORACCOUNTDISPLAYVALUE	DEFAULTDIMENSIONDISPLAYVALUE	MS Dimension combination
DEFAULTDIMENSIONSFOROFFSETACCOUNTDISPLAYVALUE	DEFAULTDIMENSIONDISPLAYVALUE	MS Dimension combination
DEPOSITNUMBER		Optional
EXCHANGERATE	EXCHANGERATE	
FINTAGDISPLAYVALUE	FINTAGDISPLAYVALUE	MS Tags Combination
ISPREPAYMENT	PREPAYMENT	
ITEMWITHHOLDINGTAXGROUP	ISWITHHOLDINGCALCULATIONENABLED	???
MARKEDINVOICE	INVOICE	Take invoiceNo from line with same (UniqueId)
MARKEDINVOICECOMPANY		M-P(Based on legal entity)
NACHAIATFOREIGNEXCHANGEINDICATOR		Optional
NACHAIATFOREIGNEXCHANGEREERENCE		Optional
NACHAIATFOREIGNEXCHANGEREERENCEINDICATOR		Optional
NACHAIATOFACSCREENINGINDICATOR		Optional
NACHAIATOFACSECONDARYSCREENINGINDICATOR		Optional
NACHAIATORIGINATINGDFIQUALIFIER		Optional
NACHAIATRECEIVINGDFIQUALIFIER		Optional
OFFSETACCOUNTDISPLAYVALUE	ACCOUNTDISPLAYVALUE (in case not cust)	Rely on settlement

D365 Field	Source (Middleware)	Notes
		on one line or more
OFFSETACCOUNTTYPE	OFFSETACCOUNTTYPE	
OFFSETCOMPANY		M-P(Based on legal entity)
OFFSETFINTAGDISPLAYVALUE		MS Tags Combination
OFFSETTRANSACTIONTEXT		Optional
OVERRIDESALESTAX		Optional
PAYMENTID		Optional
PAYMENTMETHODNAME		Optional
PAYMENTNOTES		Optional
PAYMENTREFERENCE		Optional
PAYMENTSPECIFICATION		Optional
POSTDATEDCHECKBANKBRANCH		Optional
POSTDATEDCHECKBANKNAME		Optional
POSTDATEDCHECKCASHIERDISPLAYVALUE		Optional
POSTDATEDCHECKISREPLACEMENTCHECK		Optional
POSTDATEDCHECKMATURITYDATE		Optional
POSTDATEDCHECKNUMBER		Optional
POSTDATEDCHECKORIGINALCHECKNUMBER		Optional
POSTDATEDCHECKREASONFORSTOP		Optional
POSTDATEDCHECKRECEIVEDDATE		Optional
POSTDATEDCHECKREPLACEMENTCOMMENTS		Optional
POSTDATEDCHECKSALESPERSONDISPLAYVALUE		Optional
POSTDATEDCHECKSTOPPAYMENT		Optional
POSTINGPROFILE	POSTINGPROFILE	Rely on settlement on online or more

D365 Field	Source (Middleware)	Notes
		Based on Account type
REPORTINGCURRENCYEXCHRATE	REPORTINGCURRENCYEXCHRATE	It will be set Based on MD365
REPORTINGCURRENCYEXCHRATESECONDARY	REPORTINGCURRENCYEXCHRATESECONDARY	It will be set Based on MD365
SECONDARYEXCHANGERATE	EXCHANxGERATESECONDARY	It will be set Based on MD365
SETTLEVOUCHER		MS Dynamics
TAXGROUP	SALESTAXGROUP	
TAXITEMGROUP		Optional
THIRDPARTYBANKACCOUNTID		Optional
TRANSACTIONDATE	TRANSDATE	
TRANSACTIONTEXT		Ex:Customers Collection January 2025 (Yards-Cash) Default description
USEABANKDEPOSITSLIP		Default(No)
USESALESTAXDIRECTIONFROMMAINACCOUNT		Default(No)
VOUCHER		Middleware & MS

Important Business Rules

- Invoice number is derived using **UniqueId** from the source.
- If **UniqueId** repeats:

- Multiple lines will be created.
 - Settlement will occur per customer line.
- Some customers may not have invoice numbers; these require exception handling.

Step 2: Import Customer Settlement

Middleware sends settlement data to link payments to invoices.

Key Mapping

D365 Field	Source	Notes
JOURNALBATCHNUMBER	Middleware	From step 1
JOURNALLINENUMBER	Middleware	From step 1
INVOICENUMBER	Invoice	From source
ACCOUNTDISPLAYVALUE	Customer account	
INVOICEACCOUNT	Customer account	From D365 invoice
INVOICEDUEDATE	From D365	Retrieved dynamically
SETTLEMENTAMOUNTININVOICECURRENCY	Settlement amount	From source
CASHDISCOUNTTOTAKEININVOICECURRENCY	Discount amount	From sheet
Settlement (For Dynamics)	CashIn Excel	Notes
JOURNALLINECOMPANY	M-P Default	M-P(Based on legal entity)
JOURNALBATCHNUMBER	Middleware	Created by middleware
JOURNALLINENUMBER	Middleware	Created by middleware
INVOICENUMBER	INVOICE	Middleware
INVOICECOMPANY	M-P Default	M-P(Based on legal entity) MD365
INVOICEDUEDATE	Invoice due date from Invoice in MD365f&O	Get from invoices(Dynamic)
ACCOUNTDISPLAYVALUE	ACCOUNTDISPLAYVALUE	

D365 Field	Source	Notes
CASHDISCOUNTTOTAKEININVOICECURRENCY	CASHDISCOUNTAMOUNT	Always in sheet
INVOICEACCOUNT	ACCOUNTDISPLAYVALUE	Get Customer Invoice (From Invoice dynamic)
INVOICETOPAYMENTCROSSRATE	فرق العملة بين الفاتوره و الدفع	From Invoice Dynamic
SETTLEMENTAMOUNTININVOICECURRENCY	Settle amount by Currency for Invoice	From Settlement Dynamic

Exchange Rate Logic for Currency evaluation

At posting:

- System calculates:
 - Exchange rate at **invoice date**.
 - Exchange rate at **payment date**.
- Difference is posted automatically by D365.

2) Vendor Payment Integration (Cash-Out)

Step 1: Import Vendor Payment Journal

Key Mapping (Mandatory Fields)

D365 Field	Source	Notes
JOURNALBATCHNUMBER	Middleware	Generated by middleware
LINENUMBER	Middleware	Generated by middleware
ACCOUNTDISPLAYVALUE	Vendor account	
ACCOUNTTYPE	Account type	Vendor
COMPANY	Legal entity	Based on mapping
CURRENCYCODE	Currency	

D365 Field	Source	Notes
DEBITAMOUNT / CREDITAMOUNT	Amount	
EXCHANGERATE	Exchange rate	
TRANSACTIONDATE	TransDate	
POSTINGPROFILE	Posting profile	Based on account type
MARKEDINVOICE	Invoice number	From UniqueId

Payment Line (For Dynamics)	CashOut Excel	Notes
JOURNALBATCHNUMBER	Middleware	Created by middleware
LINENUMBER	Middleware	Created by middleware
ACCOUNTDISPLAYVALUE	ACCOUNTDISPLAYVALUE	
ACCOUNTTYPE	ACCOUNTTYPE	
BANKTRANSACTIONTYPE		Optional
CALCULATEWITHHOLDINGTAX	ISWITHHOLDINGCALCULATIONENABLED	
CATEGORYPURPOSE		Optional
CENTRALBANKIMPORTDATE		Optional
CENTRALBANKPURPOSECODE		Optional
CENTRALBANKPURPOSETEXT		Optional
CHARGEBEARER		Optional
CHECKNUMBER		Optional
COMPANY		M-P(Based on legal entity)
CREDITAMOUNT	CREDITAMOUNT	
CURRENCYCODE	CURRENCYCODE	
DEBITAMOUNT	DEBITAMOUNT	
DEFAULTDIMENSIONSFORACCOUNTDISPLAYVALUE	DEFAULTDIMENSIONDISPLAYVALUE	MS Dimension combination
DEFAULTDIMENSIONSFOROFFSETACCOUNTDISPLAYVALUE	DEFAULTDIMENSIONDISPLAYVALUE	MS Dimension combination
ERRORCODEPAYMENT		Optional
EXCHANGERATE	EXCHANGERATE	
FINTAGDISPLAYVALUE	FINTAGDISPLAYVALUE	MS Tags Combination
FULLPRIMARYREMITTANCEADDRESS		Optional
ISPREPAYMENT	PREPAYMENT	
ITEMWITHHOLDINGTAXGROUP	ISWITHHOLDINGCALCULATIONENABLED	???
LOCALINSTRUMENT		Optional

MARKEDINVOICE	INVOICE	Take invoiceNo from line with same (Uniqueld)
MARKEDINVOICECOMPANY		M-P(Based on legal entity)
NACHAIATFOREIGNEXCHANGEINDICATOR		Optional
NACHAIATFOREIGNEXCHANGEREERENCE		Optional
NACHAIATFOREIGNEXCHANGEREERENCEINDICATOR		Optional
NACHAIATOFACSCREENINGINDICATOR		Optional
NACHAIATOFACSECONDARYSCREENINGINDICATOR		Optional
NACHAIATORIGINATINGDFIQUALIFIER		Optional
NACHAIATRECEIVINGDFIQUALIFIER		Optional
NEWJOURNALBATCHNUMBER		Optional
OFFSETACCOUNTDISPLAYVALUE	ACCOUNTDISPLAYVALUE (in case not vend)	Rely on settlement on oneline or more
OFFSETACCOUNTTYPE	OFFSETACCOUNTTYPE	
OFFSETCOMPANY		M-P(Based on legal entity)
OFFSETFINTAGDISPLAYVALUE	ACCOUNTDISPLAYVALUE (in case not vend)	Rely on settlement on oneline or more
OFFSETTRANSACTIONTEXT		Optional
OVERRIDESALESTAX		Optional
PAYMENTID		Optional
PAYMENTMETHODNAME		AAIB-EG-CF
PAYMENTREFERENCE		Optional
PAYMENTSPECIFICATION		Optional
POSTDATEDCHECKBANKBRANCH		Optional
POSTDATEDCHECKBANKNAME		Optional
POSTDATEDCHECKCASHIERDISPLAYVALUE		Optional
POSTDATEDCHECKISREPLACEMENTCHECK		Optional
POSTDATEDCHECKMATURITYDATE		Optional
POSTDATEDCHECKNUMBER		Optional
POSTDATEDCHECKORIGINALCHECKNUMBER		Optional
POSTDATEDCHECKREASONFORSTOP		Optional
POSTDATEDCHECKRECEIVEDDATE		Optional
POSTDATEDCHECKREPLACEMENTCOMMENTS		Optional
POSTDATEDCHECKSALESPERSONDISPLAYVALUE		Optional
POSTDATEDCHECKSTOPPAYMENT		Optional

POSTINGPROFILE	POSTINGPROFILE	Rely on settlement on online or more Based on Account type
REMITTANCEADDRESSCITY		Optional
REMITTANCEADDRESSCOUNTRY		Optional
REMITTANCEADDRESSCOUNTRYISOCODE		Optional
REMITTANCEADDRESSCOUNTY		Optional
REMITTANCEADDRESSDESCRIPTION		Optional
REMITTANCEADDRESSDISTRICTNAME		Optional
REMITTANCEADDRESSLATITUDE		Optional
REMITTANCEADDRESSLONGITUDE		Optional
REMITTANCEADDRESSSTATE		Optional
REMITTANCEADDRESSSTREET		Optional
REMITTANCEADDRESSTIMEZONE		Optional
REMITTANCEADDRESSVALIDFROM		Optional
REMITTANCEADDRESSVALIDTO		Optional
REMITTANCEADDRESSZIPCODE		Optional
REMITTANCELOCATIONID		Optional
REPORTINGCURRENCYEXCHRATE	REPORTINGCURRENCYEXCHRATE	?
REPORTINGCURRENCYEXCHRATESECONDARY	REPORTINGCURRENCYEXCHRATESE CONDARY	?
SECONDARYEXCHANGERATE	EXCHANGERATESECONDARY	?
SERVICELEVEL		Optional
SETTLEVOUCHER		MS Dynamics
TAXGROUP	SALESTAXGROUP	
TAXITEMGROUP		Optional
TAXWITHHOLDGROUP		Optional
THIRDPARTYBANKACCOUNTID		Optional
TRANSACTIONDATE	TRANSDATE	
TRANSACTIONTEXT		Customers Collection January 2025 (Yards-Cash)
USESALESTAXDIRECTIONFROMMAINACCOU NT		Default(No)
VENDORNAME		Get from invoices(Dyna mic)
VOUCHER		Middleware & MS

Notes

- Offset account depends on settlement logic.
- Vendor name is retrieved dynamically from invoice data.

Step 2: Import Vendor Settlement

Key Mapping

D365 Field		Source	Notes
JOURNALBATCHNUMBER		Middleware	
JOURNALLINENUMBER		Middleware	
INVOICENUMBER		Invoice	
ACCOUNTDISPLAYVALUE		Vendor account	
INVOICEACCOUNT		Vendor account	From D365
INVOICEDUEDATE		From D365	
SETTLEMENTAMOUNTININVOICECURRENCY		Settlement amount	
CASHDISCOUNTTOTAKEININVOICECURRENCY		Discount amount	
Settlement (For Dynamics)		CashOut Excel	Notes
JOURNALLINECOMPANY			M-P(Based on legal entity)
JOURNALBATCHNUMBER		Middleware	Created by middleware
JOURNALLINENUMBER		Middleware	Created by middleware
INVOICENUMBER		INVOICE	
INVOICECOMPANY			M-P(Based on legal entity)
INVOICEDUEDATE			Get from invoices(Dynamic)
ACCOUNTDISPLAYVALUE		ACCOUNTDISPLAYVALUE	
CASHDISCOUNTTOTAKEININVOICECURRENCY		CASHDISCOUNTAMOUNT	Always in sheet
INVOICEACCOUNT		ACCOUNTDISPLAYVALUE	Get Customer Invoice (From Invoice dynamic)
INVOICETOPAYMENTCROSSRATE			From Invoice Dynamic

SETTLEMENTAMOUNTININVOICECURRENCY		From Settlement Dynamic
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D365 Field	Middleware Field	Mandatory	Description/Rule
JOURNALBATCHNUMBER	BatchNumber	Yes	From step 1
JOURNALLINENUMBER	LineNumber	Yes	From step 1
INVOICENUMBER	InvoiceNumber	Yes	Invoice to settle
ACCOUNTDISPLAYVALUE	VendorAccount	Yes	Vendor account
INVOICEACCOUNT	VendorAccount	Yes	From D365 invoice
INVOICEDUEDATE	InvoiceDueDate	No	Retrieved from D365
SETTLEMENTAMOUNTININVOICECURRENCY	SettlementAmount	Yes	Settlement value
CASHDISCOUNTTOTAKEININVOICECURRENCY	CashDiscount	No	If applicable
CHECKNUMBER	CheckNumber	Conditional	Required when voucher type is check

Key Open Design Questions (Must Be Finalized)

- Batch name standardization**
 - Defined by Finance team in D365.
- Handling repeated UniqueId**
 - Option A: One payment line per record.

- Option B: Aggregate into one line per customer.
 - 3. **Customers without invoice numbers**
 - Define fallback logic:
 - On-account payment.
 - Suspense account.
 - Rejection.
-

Recommended Technical Architecture

Option 1: Data Management Framework (DMF) – Batch Import

Best for: High-volume, file-based integrations.

Flow:

1. Middleware generates payment file.
2. Upload to DMF entities:
 - Customer payment journal lines.
 - Customer settlement.
3. Run batch posting.

Pros

- Standard Microsoft approach.
- Low customization.
- Easier support.

Cons

- Not real-time.
-

Option 2: Custom API Integration

Best for: Real-time processing.

Flow:

1. Middleware calls custom API.
2. API creates journal lines.

3. API applies settlement.
4. Post journal.

Pros

- Real-time integration.
- Better validation control.

Cons

- Requires development and maintenance.
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Option 3: Hybrid (Recommended)

- Use API for journal creation.
- Use DMF for bulk settlements.

Rationale

- Balances performance and flexibility.
 - Reduces custom code footprint.
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Middleware Integration – Sequence Diagrams

1. API-Based Real-Time Integration

Actors:

User → Middleware → D365 F&O API → D365 Database

Sequence:

1. User triggers payment in source system
2. Middleware prepares payment payload
3. Middleware calls D365 Payment API
4. D365 validates data
5. D365 creates payment journal line
6. Middleware calls Settlement API
7. D365 retrieves open invoices

8. D365 applies settlement
9. D365 posts journal
10. D365 returns success response to middleware
11. Middleware updates source system status

2. Batch Integration Using DMF

Actors:

User → Source System → Middleware → DMF (Staging) → D365 F&O → D365 Database

Sequence:

1. User posts payments in source system
2. Source system exports payment file
3. Middleware transforms data to D365 format
4. Middleware uploads file to DMF staging entity
5. DMF validates staging data
6. DMF imports data into payment journal
7. Middleware uploads settlement file
8. DMF imports settlement data
9. Batch job posts payment journal
10. D365 applies settlement to invoices
11. Posting results stored in D365
12. Middleware retrieves status and updates source system

Next Actions

1. Confirm settlement logic for repeated UniqueId.
2. Define handling for payments without invoice numbers.
3. Agree on journal batch naming convention.
4. Choose integration approach (DMF vs API).
5. Prepare mapping specification signed off by:
 - Finance
 - Middleware team
 - D365 team

